# COMPUTER ORGANIZATION AND ARCHITECTURE

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**Goal: develop a MIPS assembly language program to perform the following tasks: Accept three integers (named x, y, and z) from the user. Calculate an integer f using the following equation:**

$$f = \frac{yz - x^2}{4} + 8x$$

- Print the value of $f$.

# step 1

```
.data
msg1:.asciiz "Enter value for y: "
msg2:.asciiz "Enter value for z: "
msg3:.asciiz "Enter value for x: "
msg4:.asciiz "the result of f : "
```

# step 2

```
.text
la $a0,msg1
li $v0,4
syscall    ##this will print msg1

li $v0,5
syscall   ## to input an integer
move $t0,$v0 ## move the frist user input into to $t0

la $a0,msg2
li $v0,4
syscall    ##this will print msg2

li $v0,5
syscall   ## to input an integer
move $t1,$v0  ## ## move the second user input into to $t1

la $a0,msg3
li $v0,4
syscall    ##this will print msg3
```

# step 3

```
li $v0,5
syscall   ## to input an integer
move $t2,$v0  #### move the third user input into to $t2

mulu $t3, $t0,$t1 ## we mulyiply the values of y,z and stor them in #t3
mulu $t4, $t2,$t2 ## we mulu the value of x by itself
                   ##to represent the value of x squared

subu $t5, $t3,$t4 ## subtraction value of t3,t4
divu $t6, $t5,4   ## divide t5 by 4
mulu $t2, $t2,8
addu $t7,$t6,$t2

la $a0, msg4
li $v0,4
syscall
```

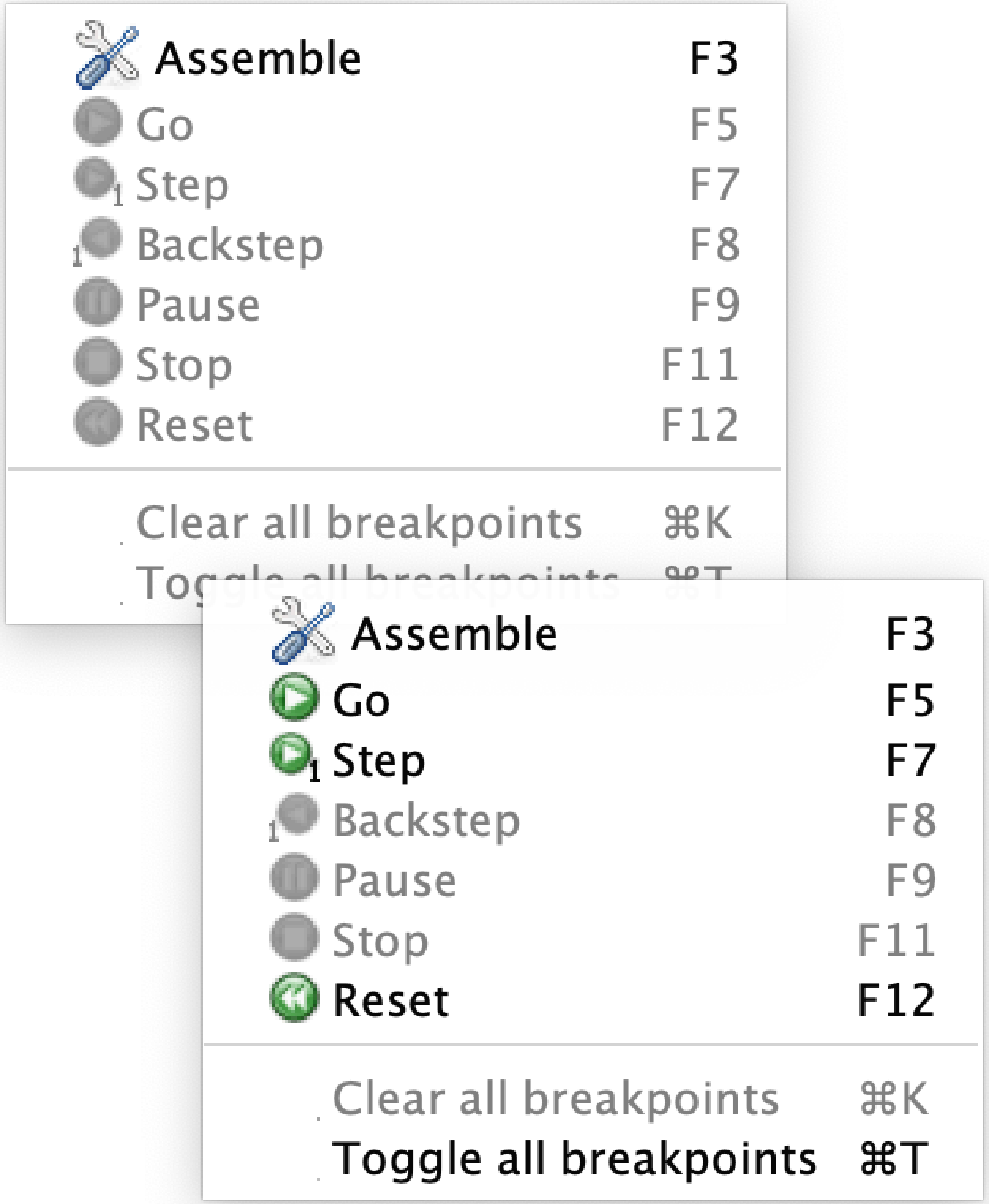
# step 4

```
move $a0, $t7
li $v0,1
syscall

li $v0,10 ## exit
syscall
```

## Steps to Run and Test the MIPS Assembly Code

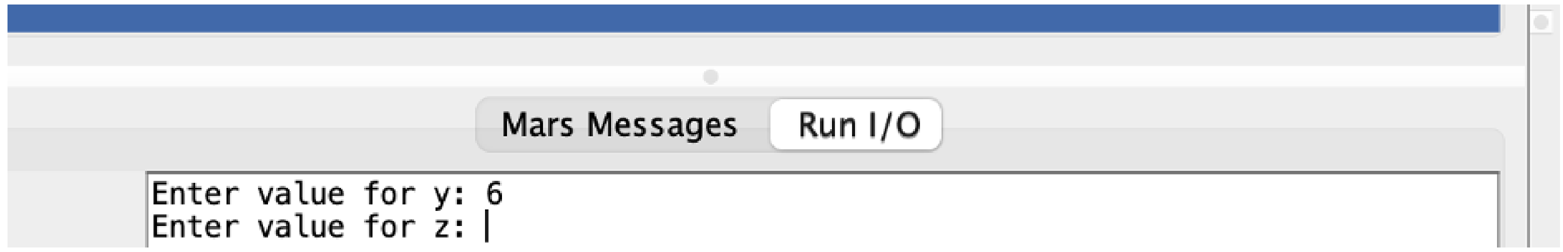
First, the “Run” menu is opened from the toolbar.
Then, “Assemble” is selected to compile the code.
After that, “Go” is clicked to start the program execution.



Once the program starts, the user is prompted to enter three integer values: x, y, and z. These values are entered one by one

1


2

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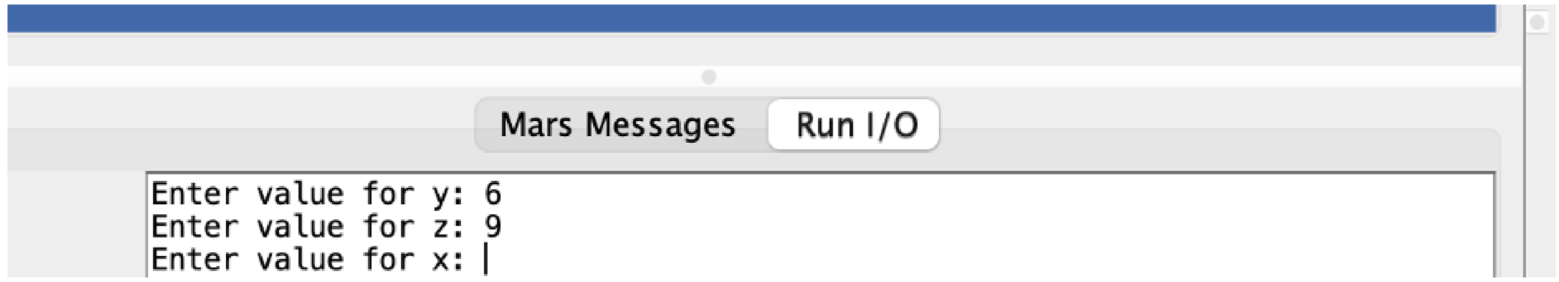
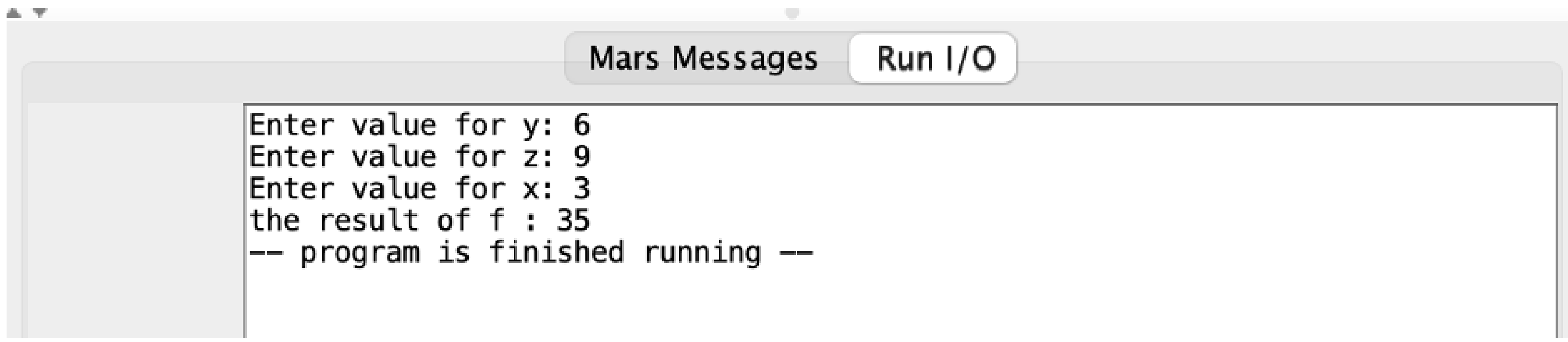


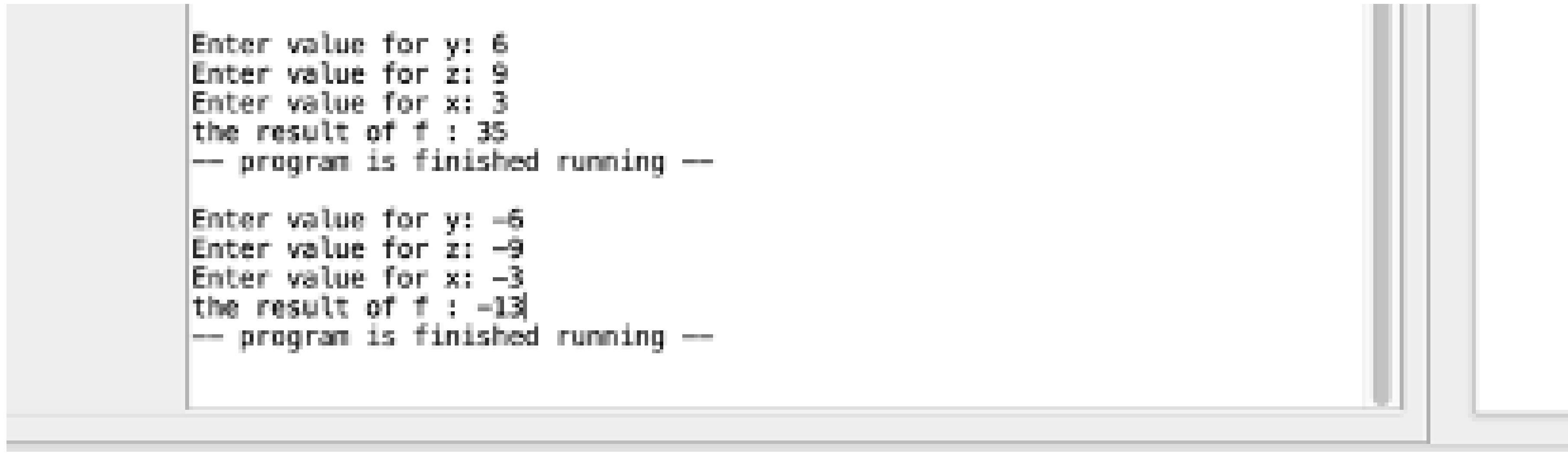
The program calculated the result based on the formula and displayed the value of f.

4




It was also tested with negative numbers.

5



# Thank you very much!